

Sebastian F. Otarola-Bustos, PhD

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Professional Summary

I am an atmospheric & hydrologic data scientist with a PhD in Atmospheric Sciences and four years of industry experience in data science and Machine Learning (ML) development for weather and hydrologic forecasting. For my PhD, I participated in two major field experiments on the U.S. west coast, where I operated scientific instrumentation to study atmospheric flows over complex terrain to improve boundary-layer parameterizations in weather models. I pursue independent research projects with a focus on operational pipelines and forecast verification, including deep learning for streamflow, climate downscaling, and live evaluation of AI and physics-based weather models.

Education

PhD, Civil & Environmental Engineering & Earth Sciences

University of Notre Dame – Notre Dame, IN | 2016 –2022

B.S., Water Resources Engineering

Pontificia Universidad Católica de Chile – Santiago, Chile | 2009 –2015

Technical Skills

Machine Learning & AI: PyTorch • scikit-learn • CNNs • LSTM • gradient boosting • diffusion models • anomaly detection • forecast verification (Brier, reliability, NSE/KGE) • bias correction • extreme value analysis • AI-assisted development

Programming & Computational Stack: Python • MATLAB • Git • SQL, SQLite, PostgreSQL • Docker • C++ • Linux • HTML • Cloud Computing • ArcGIS Pro

Atmospheric Models & Observations: NWP Forecasting Systems (GFS, ECMWF IFS/AIFS, NAM, HRRR, WRF) • CMIP6 • ERA5 • Satellite Remote Sensing (Sentinel-1, Sentinel-2, GOES) • Field instrumentation (Doppler lidar, microwave radiometer, radar, flux towers)

Research Experience

Independent Applied Research

2025 - Present

- I am training regional Long Short-Term Memory (LSTM) and diffusion models to predict streamflow in U.S. west coast basins during Atmospheric River (AR) flood events and benchmarking them against process-based and operational models (NOAA's National Water Model) across various basins and hydrological regimes. Preliminary results show the LSTM model outperforms operational forecasts on AR-day skill, and that sampling the ensemble at an upper quantile corrects peak-flow underestimation (manuscript in preparation, *Journal of Hydrology: Regional Studies*).
- I am training a convolutional neural network (CNN) to downscale CMIP6-based climate projections of extreme precipitation (Rx1day) and temperature (TXx) over Chile from $\sim 1^\circ$ to 0.05° , across a 26-model ensemble. The CNN captures Chile's unique topography, and I am investigating the physical mechanisms driving the projected changes in collaboration with researchers in Chile and the US (manuscript in preparation: *Climate Dynamics*).
- I built two weather intelligence platforms with live data ingestion of operational forecasts and observations and a bias-correction pipeline: 1) A [U.S. heat-wave tracker](#) for 165 cities with running forecast verification, and 2) [a storm tracker](#) that monitors atmospheric rivers, weather forecasts, weather station data, and flood alerts in Chile.
- During the Category 4 atmospheric river that hit Chile in July 2026, I used my storm tracker to evaluate ECMWF's AI model (AIFS) against GFS and ECMWF-IFS at rain gauges from the coast to the high Andes. Physics-based models overestimated precipitation by up to 3x at the high-Andes site, while AIFS smoothed the orographic maxima and only matched observed totals at short lead times.

University of Notre Dame – PhD Researcher, Atmospheric Fluid Dynamics

Notre Dame, IN | Aug 2016 – May 2022

- Deployed and operated flux towers, scanning Doppler lidars, and microwave radiometers in two major field campaigns, the Wind Forecast Improvement Project 2 (WFIP2) in the Columbia River Gorge, OR, and the Sundowner Wind Experiment (SWEX) in Santa Barbara, CA, and processed the observations to evaluate the skill of weather forecasting models in complex terrain.
- Published first-author results from WFIP2 in Boundary-Layer Meteorology, quantifying sub-grid variability of surface-layer turbulence in complex terrain from 18 months of tower observations and evaluating the skill of a widely used WRF boundary-layer parameterization across stability regimes.
- Designed synchronized scanning sequences for two Doppler lidars during SWEX to resolve internal gravity wave breaking ahead of Sundowner wind onset in the Santa Ynez Mountains.

Industry Experience

Hazen – Principal Scientist

Silver Spring, MD | Aug 2022 – Present

- Led machine learning (ML) initiatives including a flow forecasting tool, an urban flooding detection model, and a water demand forecasting model, integrating historical and operational data with weather forecasts.
- Developed and calibrated hydrologic models for utilities and water districts across the US, supporting flood analysis in urban and natural systems, water supply management, and evaluation of capital improvement projects.
- Contributed to a Python framework for building and automatically calibrating parsimonious (sparse-parameter) hydrological models against observed flows, aimed at wet-weather flooding in stormwater systems.
- Led data analysis and tool development for a Water Research Foundation (WRF) project on water quality in quarry lakes. The project leveraged the LAGOS dataset, which included in-situ and remote sensing measurements for about 500,000 lakes in the US. Developed a web screening tool now available to WRF subscribers.

Publications

Articles

- **Otarola-Bustos, S.F.**, & Buahin, C. (2027). LSTM and diffusion models compared with process-based and operational benchmarks during atmospheric-river floods in US West Coast basins. *Journal of Hydrology: Regional Studies*. (in preparation).
- **Otarola Bustos, S. F.**, Giani, P., & Valenzuela, R. (2027). Physical Drivers of Downscaled CMIP6 Projections of weather extremes over Chile. *Climate Dynamics* (in preparation).
- **Otarola-Bustos, S. F.** et al. (2023). Subgrid variability of surface-layer parameters in complex terrain. *Boundary-Layer Meteorology*, 187, 229–265.
- Grachev, A. et al. (2022). Hybrid bulk algorithm for turbulent fluxes over dry and wet soils. *Journal of Applied Meteorology and Climatology*, 61, 393–414.
- Grachev, A. et al. (2020). On the surface energy balance closure at different temporal scales, *Agricultural and Forest Meteorology*.
- Wilczak, J. M. et al. (2019); Fernando, H. J. S. et al. (2019). WFIP2 and Perdígão field campaign overviews. *Bulletin of the American Meteorological Society*.
- Fernando, H. J. et al. (2018). The Perdígao: Peering into microscale details of mountain winds, *B. Am. Meteorol. Soc.*, 100, 799–820.
- Wilcox, A. C. et al. (2016). Integrated analysis of the March 2015 Atacama floods. *Geophysical Research Letters*, 43, 8035–8043.

Conferences

- **Otarola Bustos, S. F.**, Giani, P., & Valenzuela, R. (2026). Downscaling CMIP6 projections of weather extremes over Chile: A CNN super-resolution approach. Abstract submitted to Session A060: Decision-Relevant Understanding of Impactful Weather and Extremes, AGU Fall Meeting 2026, San Francisco, CA, 7–11 Dec.
- **Otarola Bustos, S. F.**, & Buahin, C. (2026). LSTM and diffusion models compared with process-based and operational benchmarks during atmospheric-river floods in US West Coast basins. Abstract submitted to Session H039: AI-Driven Hydrologic Modeling and Prediction: Methods, Applications, and Emerging Frontiers, AGU Fall Meeting 2026, San Francisco, CA, 7–11 Dec.
- **Otarola Bustos, S. F.** Measurements of surface layer parameters in complex terrain during the WFIP experiment in the Columbia River Gorge (OR). AGU Fall Meeting 2022, Chicago, IL, 12-16 Dec.
- Duine, G. J., Carvalho, L. M. V., De Wekker, S. F. J., Jones, C., de Orla-Barile, M., Whang, Z., Kasić, J., Fernando, H. J. S., & **Otarola Bustos, S. F.** (2022). Exploring the spatiotemporal variability of lee slope jets during the Sundowner Winds Experiment. 20th Conference on Mountain Meteorology, American Meteorological Society, Park City, UT, June 2022.

Teaching & Mentoring

Engineering Mentoring – Hazen and Sawyer

Aug 2022 – Present

- Mentored junior engineers in hydraulic modeling (InfoWater Pro, PCSWMM), GIS software, and Python-based workflows to automate labor-intensive tasks and increase work accuracy.
- Trained junior engineers and scientist in GitHub workflows and best development practices in collaborative data science and machine learning projects.

Graduate Teaching Assistant – University of Notre Dame

2017 – 2018

- **CE 30455: Environmental Hydrology** (Fall 2017, Fall 2018). Conducted office hours and problem-solving sessions ahead of mid-terms. Graded homework's, mid-terms, and exams. Prepared laboratory demonstrations (Prof. Andrew Kennedy).
- **CE 30460: Fluid Mechanics** (Spring 2017, Spring 2018). Conducted office hours and problem-solving sessions ahead of mid-terms. Graded homework's, mid-terms, and exams (Prof. David Richter).